

Artificial Intelligence

AI2002

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EVALUATION MEASURES

CLASSIFICATION

Confusion Matrix

- A confusion matrix for a two classes (+, -) is shown below.

	C ₁	C ₂
C ₁	True positive	False negative
C ₂	False positive	True negative

	+	-
+	++	+-
-	-+	--

- There are four quadrants in the confusion matrix, which are symbolized as below.
 - True Positive** (TP: f_{++}) : The number of instances that were positive (+) and correctly classified as positive (+).
 - False Negative** (FN: f_{+-}): The number of instances that were positive (+) and incorrectly classified as negative (-).
 - False Positive** (FP: f_{-+}): The number of instances that were negative (-) and incorrectly classified as (+).
 - True Negative** (TN: f_{--}): The number of instances that were negative (-) and correctly classified as (-).

Confusion Matrix

Note:

- $N_p = TP(f_{++}) + FN(f_{+-})$
= is the total number of positive instances.
- $N_n = FP(f_{-+}) + TN(f_{--})$
= is the total number of negative instances.
- $N = N_p + N_n$
= is the total number of instances.
- $(TP + TN)$ denotes the number of correct classification
- $(FP + FN)$ denotes the number of errors in classification.
- For a perfect classifier, $FP = FN = 0$

Confusion Matrix Example

- For example,

Class	+	-
+	52 (TP)	18 (FN)
-	21 (FP)	123 (TN)

Calculate the performance evaluation metrics

Accuracy

- It is defined as the fraction of the number of examples that are correctly classified by the classifier to the total number of instances.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

Performance Evaluation Metrics

- We now define a number of metrics for the measurement of a classifier.
 - In our discussion, we shall make the assumptions that there are only two classes: + (positive) and – (negative)
- **True Positive Rate (TPR)**: It is defined as the fraction of the positive examples predicted correctly by the classifier.

$$TPR = \frac{TP}{P} = \frac{TP}{TP+FN} = \frac{f_{++}}{f_{++}+f_{+-}}$$

- This metrics is also known as *Recall*, *Sensitivity* or *Hit rate*.
- **False Positive Rate (FPR)**: It is defined as the fraction of negative examples classified as positive class by the classifier.

$$FPR = \frac{FP}{N} = \frac{FP}{FP + TN} = \frac{f_{-+}}{f_{-+}+f_{--}}$$

Performance Evaluation Metrics

- **False Negative Rate (FNR):** It is defined as the fraction of positive examples classified as a negative class by the classifier.

$$FNR = \frac{FN}{P} = \frac{FN}{TP + FN} = \frac{f_{+-}}{f_{++} + f_{+-}}$$

- **True Negative Rate (TNR):** It is defined as the fraction of negative examples classified correctly by the classifier

$$TNR = \frac{TN}{N} = \frac{TN}{TN + FP} = \frac{f_{--}}{f_{--} + f_{-+}}$$

- This metric is also known as *Specificity*.

Performance Evaluation Metrics

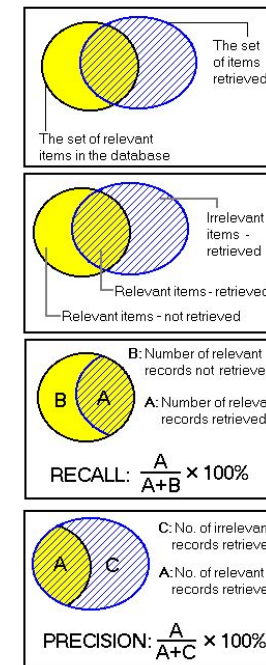
- Both, **Precision** and **Recall** are defined by

- $$Precision = \frac{TP}{TP+FP}$$

- $$Recall = \frac{TP}{TP+FN}$$

Recall and Precision

- Recall:** The percentage of the total relevant documents in a database retrieved by your search.
- If you knew that there were 1000 relevant documents in a database and your search retrieved 100 of these relevant documents, your recall would be 10%.
- Precision:** The percentage of relevant documents in relation to the number of documents retrieved.
- If your search retrieves 100 documents and 20 of these are relevant, your precision is 20%.



Performance Evaluation Metrics

- **F₁ Score (F₁):** Recall (r) and Precision (p) are two widely used metrics employed in analysis..
 - It is defined in terms of (r or Recall) and (p or Precision) as follows.

$$F_1Score = \frac{2 * Recall . Precision}{Recall + Precision} = \frac{2TP}{2TP + FP + FN}$$

Note

- F₁ represents the harmonic mean between recall and precision
- High value of F₁ score ensures that both Precision and Recall are reasonably high.

Analysis with Performance Measurement Metrics

- Based on the various performance metrics, we can characterize a classifier.
- We do it in terms of TPR, FPR, Precision and Recall and Accuracy
- **Case 1: Perfect Classifier**

When every instance is **correctly** classified, it is called the **perfect classifier**. In this case, $TP = P$, $TN = N$ and CM is

$$TPR = TP/(TP+FN) = \frac{P}{P} = 1$$

$$FPR = \frac{0}{N} = 0$$

$$Precision = \frac{P}{P} = 1$$

$$F_1 \text{ Score} = \frac{2 \times 1}{1+1} = 1$$

$$Accuracy = \frac{P+N}{P+N} = 1$$

		Predicted Class	
		+	-
Actual class	+	P	0
	-	0	N

Analysis with Performance Measurement Metrics

- **Case 2: Worst Classifier**

When every instance is **wrongly** classified, it is called the **worst classifier**. In this case, $TP = 0$, $TN = 0$ and the CM is

$$TPR = \frac{0}{P} = 0$$

$$FPR = \frac{N}{N} = 1$$

$$Precision = \frac{0}{N} = 0$$

F_1 Score = Not applicable
as $Recall + Precision = 0$

$$Accuracy = \frac{0}{P+N} = 0$$

		Predicted Class	
		+	-
Actual class	+	0	P
	-	N	0

Analysis with Performance Measurement Metrics

- **Case 3: Ultra-Liberal Classifier**

The classifier always predicts the + class correctly. Here, the False Negative (FN) and True Negative (TN) are zero. The CM is

$$TPR = \frac{P}{P} = 1$$

$$FPR = \frac{N}{N} = 1$$

$$Precision = \frac{P}{P+N}$$

$$F_1 \text{ Score} = \frac{2P}{2P+N}$$

$$Accuracy = \frac{P}{P+N} = 0$$

		Predicted Class	
		+	-
Actual class	+	P	0
	-	N	0

Analysis with Performance Measurement Metrics

- **Case 4: Ultra-Conservative Classifier**

This classifier always predicts the - class correctly. Here, the False Negative (FN) and True Negative (TN) are zero. The CM is

$$TPR = \frac{0}{P} = 0$$

$$FPR = \frac{0}{N} = 0$$

Precision = Not applicable
(as $TP + FP = 0$)

F₁ Score = Not applicable

$$\text{Accuracy} = \frac{N}{P+N} = 0$$

		Predicted Class	
		+	-
Actual class	+	0	p
	-	0	N